

Rank-3 fusion rings

A reproducible census through multiplicity 1000

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Abstract

We explain the exact enumeration of all based fusion rings of rank 3 through multiplicity 1000, comprising 460,353 isomorphism classes. The account separates the mathematical coverage argument, the completed search, and independent verification of the emitted tensors. All duality types are included, and no categorifiability filter is applied. Code, machine-readable counts, full multiplication tensors and provenance records accompany the text.

AI assistance and verification. GPT-6 Astra Pro assisted with code, exposition and computational checks. Exhaustive-search arguments and exact independent tensor verification are recorded separately; no proof-assistant certification or peer-reviewed acceptance is claimed.

Scope. A census of based rings, not a classification of tensor categories. Only completed whole-rank results are released.

Contents

1	Objects, equivalence and the counting convention	3
1.1	Why duality types can be searched separately	3
2	The certified release at rank 3	3
3	The classical rank-three Diophantine parameterization	4
3.1	Every boundary and every arithmetic progression	4
3.2	The deliberate endpoint	4
4	What the independent verification establishes	4
4.1	Provenance and reproducibility	5
4.2	Computation budgets and release gates	5
5	Reading and reusing the data	5
6	Reproduction commands	6

1 Objects, equivalence and the counting convention

A *fusion ring* here is a free abelian group with a distinguished finite basis $B = \{b_0, b_1, \dots, b_{r-1}\}$, an associative multiplication with unit b_0 , and an involution $i \mapsto i^*$ fixing 0. Its structure constants satisfy

$$b_i b_j = \sum_k N_{ij}^k b_k, \quad N_{ij}^k \in \mathbb{Z}_{\geq 0},$$

$$N_{0j}^k = N_{j0}^k = \delta_{jk}, \quad N_{ij}^0 = \delta_{i^*, j}, \quad (1)$$

$$N_{ij}^k = N_{i^* j^*}^k = N_{kj^*}^i. \quad (2)$$

These conventions agree with the usual based-ring definition; see [1, 2]. No assumption of categorifiability, commutativity or integrality of Frobenius–Perron dimensions is imposed unless proved in the relevant rank.

A based isomorphism is a bijection of the distinguished bases preserving the unit and multiplication. In tensor notation it is a permutation q fixing 0 such that

$$N_{ij}^k = N_{q(i)q(j)}^{q(k)} \quad \text{for every } i, j, k.$$

The coefficient of the unit determines the involution, so such an isomorphism automatically intertwines duality. The *multiplicity* is the maximum entry of the *full* ordered multiplication tensor. It is at least one, including for group rings. Put

$$c_r(m) = \#\{\text{rank-}r \text{ fusion rings of multiplicity exactly } m\} / \cong, \quad C_r(M) = \sum_{m=1}^M c_r(m).$$

A list through M contains all rings of multiplicity at most M , not only those of multiplicity equal to M . This distinction is essential when exporting OEIS b-files.

1.1 Why duality types can be searched separately

An involution on the $r - 1$ nonunit elements consists of p disjoint transpositions and $r - 1 - 2p$ fixed points, with $0 \leq p \leq \lfloor (r - 1)/2 \rfloor$. Fix one representative d_p of each type. An isomorphism between two tensors with involution d_p commutes with d_p . Hence it is sufficient to quotient that stratum by the centralizer

$$Z_{S_{r-1}}(d_p), \quad |Z_{S_{r-1}}(d_p)| = 2^p p! (r - 1 - 2p)!.$$

Different p cannot be isomorphic. Therefore a whole-rank census is complete exactly when every required stratum has been searched exhaustively. Finishing only the non-self-dual strata does not certify a whole-rank counting term.

2 The certified release at rank 3

The released bound is $M = 1000$ and the cumulative number of classes is $C_3(1000) = 460,353$. Every duality type is included. The accompanying full tensor file has one record per based-isomorphism class.

The first sixteen terms are 4, 3, 4, 6, 5, 9, 6, 10, 12, 9, 10, 20, 9, 13, 16, 25. The complete 1,000-term file is supplied as `oeis/b354471.txt`.

The complete-run evidence is available through `results/census.json` and the corresponding reports under `verification/`. Earlier ranks retain their original completed-run provenance; repository checks and newly executed runs are labelled separately.

3 The classical rank-three Diophantine parameterization

This companion does not claim a new theoretical rank-three classification. It materializes the familiar parameterization through the requested bound. In the self-dual basis $(1, X, Y)$, reciprocity gives

$$X^2 = 1 + mX + kY, \quad XY = kX + lY, \quad Y^2 = 1 + lX + nY.$$

Comparing $(X^2)Y$ with $X(XY)$ proves that associativity is equivalent to

$$k^2 + l^2 = 1 + lm + kn. \quad (3)$$

Exchanging X, Y sends (k, l, m, n) to (l, k, n, m) . Retain $k < l$, or $k = l$ with $m \leq n$.

If $X^* = Y$, reciprocity instead gives

$$X^2 = aX + bY, \quad Y^2 = bX + aY, \quad XY = YX = 1 + aX + aY.$$

Associativity implies $b^2 - a^2 = 1$. Nonnegative integers force $a = 0$, $b = 1$, which is exactly $\mathbb{Z}[C_3]$.

3.1 Every boundary and every arithmetic progression

For the self-dual part take $k \leq l$. If $k = 0$, equation (3) gives $l = 1$, $m = 0$, with n arbitrary in $[0, M]$. If $k = l$, it gives $k = l = 1$ and $m + n = 1$, leaving the representative $(1, 1, 0, 1)$. For $0 < k < l$, a common divisor of k, l would divide one, so $\gcd(k, l) = 1$. Set $C = k^2 + l^2 - 1$ and choose $0 \leq m_0 < k$ with $lm_0 \equiv C \pmod{k}$. For $k = 1$ take $m_0 = 0$. Then every integral solution is

$$m = m_0 + kt, \quad n = n_0 - lt, \quad n_0 = (C - lm_0)/k,$$

where

$$\max\left(0, \left\lceil \frac{n_0 - M}{l} \right\rceil\right) \leq t \leq \min\left(\left\lfloor \frac{M - m_0}{k} \right\rfloor, \left\lfloor \frac{n_0}{l} \right\rfloor\right).$$

The Euclidean algorithm finds the inverse of l modulo k . This visits every bounded solution exactly once. The two boundary cases and $\mathbb{Z}[C_3]$ complete the census. Every emitted solution is checked again in (3); the independent tensor verifier does not use that equation.

3.2 The deliberate endpoint

The repository hard-caps the rank-three generator at $M = 1000$. It contains all exact-multiplicity counts and all multiplication tensors through that bound. There are no rank-three counts or searches beyond that endpoint in the released data. The b-file belongs to OEIS A354471.

4 What the independent verification establishes

The enumerator and verifier have separate mathematical tasks. Exhaustiveness uses the complete coordinate parameterization, the safe-pruning proof and the normal completion of every required branch. The verifier reads only the emitted *full tensors*; it neither reconstructs the parameterization nor trusts the generator's polynomial list.

For every tensor it checks nonnegative integral coefficients, both unit laws, uniqueness and involutivity of the dual, both reciprocity identities, and every integer associativity identity for all four indices. It then canonicalizes under *every* unit-fixing basis permutation. A matching hash is only a lookup aid: actual canonical tensors are compared. This detects both literal repetitions and different labellings of an isomorphic based ring.

Passing these checks proves soundness and uniqueness of the emitted data; by itself it would not prove that no ring was omitted. That is why normal completion logs and the mathematical coverage argument are retained separately. Comparison with the original ancillary file is an additional containment/regression test, not an input to enumeration and not a replacement for an exhaustiveness proof.

4.1 Provenance and reproducibility

The retained Vercleyen–Slingerland v4 ancillary file has SHA-256

`b587c3f683157369da7cc090f762bbff0e18dcf11acc3f5031d6bc0c643843f8`

Its 28,451 records contain 25,138 distinct classes. The 3,313 redundant records occur at rank five and exact multiplicities 9–12. Their original and corrected counts are respectively

$$(1463, 1794, 2283, 3049) \longrightarrow (863, 1082, 1383, 1948).$$

Every deletion has an explicit basis-permutation witness in the audit folder. This audit does not establish completeness of the source’s partial-search cells.

The primary manifest `results/census.json` records every released count, bound, filename and checksum. Compressed tensors use the original line-per-ring, nested-curly-brace format. An entry is $N[i][j][k] = N_{ij}^k$: the output index has already been dualized where the internal parameter convention requires it. The raw tensor needs no additional index conversion.

4.2 Computation budgets and release gates

The private repository provides two manually dispatched GitHub Actions campaigns. The original frontier has a 1,100-second shared driver deadline and a 20-minute job ceiling. The long frontier selects only ranks 5–8: each rank has one 4,200-second (70-minute) shared deadline and a 75-minute job ceiling, at most 300 runner-minutes in total. Compilation, all attempted multiplicities and duality types, export, compression and independent verification share the driver deadline. Enumeration receives 90% of remaining driver time; the balance is reserved for other phases. Threads follow the actual runner CPU allocation.

Only a complete whole-rank run with successful independent verification enters `best/`. A later incomplete attempt never replaces it. Candidates require explicit review and fresh independent verification before integration. Partial logs remain diagnostic evidence and never supply exhaustive census or OEIS terms. All workflows are manual; neither a push nor this manuscript launches a search.

The reviewed long campaign, GitHub run 35440299307 at source revision 931addc686e8 (full revision in the review record), completed rank five through multiplicity 19 with 34,133 classes, including 5,640 at exact multiplicity 19. The subsequent rank-five bound 20 and the attempted bounds 7, 4 and 2 at ranks six, seven and eight exhausted their search budgets. They yield no larger complete whole-rank counts. Ranks three and four were not recomputed. The accepted rank-five data underwent a fresh exhaustive tensor and based-isomorphism audit on GitHub before integration. The full earlier exact-multiplicity prefix agrees. See `docs/CAMPAIGN_REVIEW.md` and `verification/imports/rank5_through19/`.

5 Reading and reusing the data

The result files count based rings, not monoidal categories or realizations of a ring. One tensor can admit several inequivalent categorifications, or none. No number here is a classification of fusion

categories. The rank-specific count files use exact multiplicity; cumulative totals belong only in the summary column. Proposed OEIS additions must be regenerated from a complete manifest, not copied from an interrupted log.

The retained code was generated with AI assistance and is accompanied by explicit mathematical coverage arguments and independent integer checks. These checks are not a formal proof in a proof assistant. The present text is an explanatory computational companion, not a claim of journal acceptance or external peer review.

6 Reproduction commands

From the repository root, a complete run can be requested as follows. A timeout reports failure to finish, not a smaller theorem.

```
python3 code/run_census.py --rank 3 --bound 1000 \
  --seconds 1200 --threads "$(nproc)" --verify --out rerun_rank3
```

To verify the supplied release without recomputing it:

```
python3 scripts/check_release.py --full
```

For a shared-budget frontier search, use `scripts/extend_census.py`. For ranks five through eight it starts at the released bound plus one and restarts the unfinished bound without claiming checkpoint resumption. The low-level command above limits enumeration only; the frontier wrapper additionally bounds every phase. Rank three is instead capped at 1000.

References

- [1] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015.
- [2] G. Vercleyen and J. Slingerland, *On Low Rank Fusion Rings*, Journal of Mathematical Physics 64 (2023), 091703; [arXiv:2205.15637v4](#).
- [3] V. Ostrik, *On formal codegrees of fusion categories*, Mathematical Research Letters 16 (2009), 895–901; [arXiv:0810.3242](#).
- [4] J. Dong and S. Palcoux, rank-four and rank-five census derivations, project manuscripts and computational supplements, September 2026. The relevant derivations are retained in the repository's `methods/` directory.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A348305, A354471, A354472, A354473, A354475, A354476 and A354477. Proposed updates in this repository are not submitted edits.